



Original Research Article

Ultra-processed food consumption is associated with the acceleration of biological aging in the Moli-sani Study



Simona Esposito¹, Alessandro Gialluisi^{1,2}, Augusto Di Castelnuovo¹, Simona Costanzo¹, Antonietta Pepe¹, Emilia Ruggiero¹, Amalia De Curtis¹, Mariarosaria Persichillo¹, Chiara Cerletti¹, Maria Benedetta Donati¹, Giovanni de Gaetano¹, Licia Iacoviello^{1,2,*}, Marialaura Bonaccio¹, for the Moli-sani Study Investigators

¹ Research Unit of Epidemiology and Prevention, IRCCS Neuromed, Pozzilli IS, Italy; ² Department of Medicine and Surgery, LUM University, Casamassima BA, Italy

ABSTRACT

Background: Healthy diets have been inversely associated with biological aging. However, the nutritional content is only one aspect of the overall food health potential, and more recently, increasing attention has been paid to nonnutrient food characteristics, such as food processing.

Objectives: To examine the association of ultra-processed food (UPF) consumption with biological aging measured by circulating blood biomarkers.

Methods: Cross-sectional analyses were conducted on 22,495 participants enrolled in the Moli-sani Study (2005–2010, Italy). Food intake was assessed by a 188-item food frequency questionnaire. UPF was defined according to the Nova classification and calculated as the ratio (weight ratio; %) between UPF (g/d) and total food eaten (g/d). Diet quality was assessed by the Mediterranean Diet Score (MDS; ranging 0–9). A deep neural network approach based on 36 circulating biomarkers was used to compute biological age (BA), and the resulting difference (Δ age = BA – chronological age)—an index of biological aging—was tested as dependent variable in multivariable linear regression analyses including known risk factors.

Results: The mean Δ age in this population was -0.70 (standard deviation ± 7.70) years. In multivariable-adjusted analyses, higher intake of UPF was associated with accelerated Δ age [$\beta = 0.34$ years; 95% confidence interval (CI): 0.08, 0.61 for the last fifth compared with the first]. This association was not linear (P value for overall association < 0.001 ; P value for nonlinearity = 0.049). Inclusion of the MDS into the model slightly attenuated this association by 9.1% ($\beta = 0.31$ years; 95% CI: 0.04, 0.59).

Conclusions: A diet rich in UPF was associated with an acceleration of biological aging in a large sample of Italian adults. The poor nutritional composition of highly processed foods weakly accounted for this association, suggesting that biological aging could be adversely influenced by non-nutrient characteristics of these foods.

Keywords: ultra-processed foods, biological aging, diet quality, accelerated aging, blood-based measure of biological age

Introduction

The prolonged survival of the population has prompted interest in understanding the origins, mechanisms and processes of aging, as well as seeking methods to mitigate, delay, or even reverse its effects [1]. However, chronological aging merely refers to the flow of time, but may not reflect the actual health status of an individual. For this reason, there has been an increasing attention to the concept of biological aging, which indicates the functional or physiological aging of an

organism and relates to functional decline [2,3]. Several mechanisms have been implicated in biological aging, both at the organism and the organ level. Among these, the most prominent include inflammation, lipid and carbohydrate transport, metabolism, and homeostasis, suggesting them as potential molecular targets for future antiaging interventions [4]. This, along with evidence of implication of other contributors, like kidney and liver functions, makes blood markers a promising and cost-effective substrate to estimate biological aging of an organism.

Abbreviations: BA, biological age; CA, chronological age; CI, confidence interval; CRP, C-reactive protein; CV, coefficient of variability; CVD, cardiovascular disease; DASH, Dietary Approach to Stop Hypertension; DNN, deep neural network; FFQ, food frequency questionnaire; FSAm-NPS, modified Food Standards Agency Nutrient Profiling System; MDS, Mediterranean Diet Score; QC, quality control; UPF, ultra-processed food; WBC, white blood cell.

* Corresponding author.

E-mail address: licia.iacoviello@moli-sani.org (L. Iacoviello).

<https://doi.org/10.1016/j.ajcnut.2024.10.006>

Received 28 February 2024; Received in revised form 4 October 2024; Accepted 9 October 2024; Available online 4 November 2024

0002-9165/© 2024 American Society for Nutrition. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Indeed, individuals of the same chronological age (CA) may exhibit different susceptibilities to age-related diseases and death, which is likely reflecting differences in their underlying biological aging processes [2, 5]. It is estimated that only a small proportion of variation in aging and human longevity is due to genetic factors [6], whereas a more prominent role is played by lifestyles—including diet—and other environmental (that is, nongenetic) factors [7]. This distinction is critical especially for aging people, and even more in light of the progressive aging of the world's population [8]. Healthy nutritious diets, such as the Mediterranean Diet and the Dietary Approach to Stop Hypertension (DASH) diet, have been favorably associated with several hallmarks of aging, including cognitive impairment, telomere length, and DNA methylation in numerous cohorts [9,10]. Consistently, proinflammatory diets (for example, typically rich in processed meat, added sugar, and refined grains) have been related to accelerated biological aging [11].

More recently, the evaluation of an individual's diet exclusively in terms of nutritional composition has been questioned, because the nutritional content may not exhaustively reflect the overall food health potential [12] and this could have varying effects on human health [13]. For this reason, it has been proposed to pay increasing attention to nonnutrient food characteristics, such as food processing [14]. The Nova classification has been conceived and developed with the purpose of accounting for the degree of food processing regardless of their nutrient content, based on the hypothesis that the extent and purpose of food processing, rather than the nutritional composition alone, could have varying effects on human health [13]. Ultra-processed foods (UPFs) are mostly industrial formulations designed to maximize palatability and consumption through a combination of calorie-dense ingredients and chemical additives with a cosmetic function, and have been associated with increased risk of mortality and noncommunicable diseases in numerous cohorts [13,14]. UPFs tend to have a poor nutritional quality, being high in sugar, salt, saturated fats but low in fiber, minerals, and vitamins, although their harmful impact on human health is likely to be at least partly independent of their low nutritional content [15].

The relationship between biological aging and UPF has been poorly explored so far. Analyses in an elderly Spanish population showed that high consumption of UPFs was associated with shorter telomere length, which is a measure of biological age (BA) [16]. In line with this report, an inverse relationship between high UPF consumption and cognitive performance, an index of healthy aging, was found in older adults from other populations [17], and a larger dietary share of UPFs has been associated with cognitive decline among Brazilian adults [18]. Eating more UPFs reportedly leads to an increased risk of frailty, another healthy aging index [19].

We therefore sought to examine the cross-sectional association of UPF consumption with a blood-based measure of biological aging, in a general population of adults recruited from Central-Southern Italy within the Moli-sani Study; as a secondary aim, we tested the role of diet quality as a potential explanatory factor in the case of an observed association.

Methods

Study population

We analyzed data from the Moli-sani Study, a large population-based cohort study designed to investigate genetic and environmental risk factors in the onset of cardiovascular, cerebrovascular, and tumor diseases. Between 2005 and 2010, a total of 24,325 subjects (aged ≥ 35 y, 52.0% females) were randomly recruited from city-hall registries of the Molise region, in the Central-Southern part of Italy. Exclusion criteria were pregnancy at the time of recruitment, disturbances in

mental or decision-making impairments, current poly-traumas or coma, or refusal to sign the informed consent form. In this cohort, response rate to recruitment invitation was 70%, that is substantially higher than in many other cohort studies [20,21].

Participants who refused to participate were older, and had higher prevalence chronic health conditions than those accepting to participate. Additional details of the study design are available elsewhere [22]. The Moli-sani Study complies with the Declaration of Helsinki and was granted the approval of the Ethics Committee of the Catholic University in Rome, Italy.

Blood sample collection, computation of BA, and selection of the analytic sample

Blood samples were collected at baseline (2005–2010) in participants who had fasted overnight and had refrained from smoking for ≥ 6 h. Lipids (total cholesterol, HDL-cholesterol, triglycerides) and blood glucose were assayed in serum samples by enzymatic reaction methods using an automatic analyzer (ILab 350, Instrumentation Laboratory, Milan, Italy) and quality control (QC) for lipids and glucose was obtained by 2 commercial standards SeraChem 1 (a control for normal levels) and SeraChem 2 (a control for pathologically high levels). The coefficients of variability (CV) of these 2 commercial standards were respectively 4.9% and 5.2% for blood total cholesterol, 3.2% and 3% for HDL-cholesterol, 5.2% and 5.3% for triglycerides, and 4.7% and 4.1% for blood glucose.

High-sensitivity C-reactive protein (CRP) was measured in fresh serum samples by a particle-enhanced immune-turbidimetric assay (ILab 350, Instrumentation Laboratory). QC for CRP was maintained using in-house serum pool and commercial laboratory standard; inter-day CV for CRP were 5.5% and 4.2%.

Hemocromocytometric analysis was performed by cell count (Coulter HMX, Beckman Coulter) within 3 h from blood collection. QC was performed by using 3 different levels of standards: 1) Abnormal 1 a pathologically high control (Abn1); 2) Abnormal 2 (Abn2) a pathologically low control; and 3) Normal (Coulter HMX, Beckman Coulter). CV for white blood cells (WBCs) was 6.2%, 3.3%, and 3.0% for Abn1, Abn2, and Normal, respectively.

Apolipoprotein A1 (ApoA), apolipoprotein B100 (ApoB100), Lipoprotein a (Lp(a)), markers of renal function (Cystatin-C, Creatinine), insulin, C-peptide, and serum vitamin D were measured subsequently on thawed samples stored frozen in liquid nitrogen at the Neuromed Biobanking Center. This was performed in the framework of the collaborative BiomarCaRE research project (EUPF7, HEALTH-F2-2011-278913), whose primary objective was to assess the value of both established and emerging biomarkers for cardiovascular disease (CVD) risk prediction by using data from 23 cohorts across Europe [23].

Computation of BA and selection of the analytic sample

To compute BA, we used a supervised machine learning algorithm called deep neural network (DNN) built in R (<https://www.r-project.org/>), through the Keras package (<https://cran.r-project.org/web/packages/keras/index.html>).

We first carried out a QC and elaboration of available data ($N = 24,325$), as briefly described below. Unreliable blood markers levels—that is, leukocyte counts whose fractions summed to either $<99\%$ or $>101\%$ —were set to missing, and missing values were imputed through either a k-nearest neighbor algorithm or a minimum/maximum imputation, see Gialluisi et al. [7] for details. Composite variables like total cholesterol, plateletcrit, hematocrit, and mean corpuscular

hemoglobin were removed to avoid collinearity. Non-Caucasian participants and/or participants reporting a nonfasting state at the time of blood draw were removed from the sample. This left 23,858 participants passing QC, with 36 circulating markers available for analysis, including lipid biomarkers (triglycerides, HDL and LDL-cholesterol, lipoprotein a, apolipoprotein A1 and B), markers of glucose metabolism (glucose, C-peptide and insulin); liver enzymes (aspartate transaminase and alanine aminotransferase), cardiac and vascular markers (NT-proB-type Natriuretic Peptide and high-sensitivity cardiac troponin I), hormones (testosterone and vitamin D), hemostasis markers such as D-Dimer, renal markers (uric acid, albumin, creatinine, cystatin-C), inflammation markers like high-sensitivity CRP; common hemochrome markers (red blood cell count, distribution width, hematocrit, hemoglobin levels, mean corpuscular volume, mean corpuscular hemoglobin concentration, total WBCs, lymphocytes, monocytes, granulocytes, neutrophils, basophils and eosinophils, platelet count, mean platelet volume, and platelet distribution width). These variables underwent normalization before analysis and were used as input features of the DNN for the prediction of BA, along with recruiting center and sex of each participant, using CA as a label.

In the original work where the DNN algorithm was deployed [4], we split the available QCed dataset ($N = 23,858$) into a random training set—a portion of the cohort (80%, $N = 19,086$) where the DNN algorithm had known labels (CA) and was trained over 1000 epochs to minimize mean squared error between CA and BA (loss function)—and a test set—remaining 20% of the cohort, $N = 4772$, where the label is initially unknown and is used to validate the deployed algorithm. In the test set, the BA measure resulting from the optimized DNN showed a mean absolute error of 6.00 y compared with CA, Pearson's correlation (r) of 0.76 and Nagelke's coefficient of determination (R^2) of 0.57. Details on QC, DNN architecture, and features used are reported in Gialluisi et al. [4]. For the purpose of the present analysis, we used BA for the totality of samples passing QC ($N = 23,858$). We then excluded individuals with missing data on diet ($n = 97$), those reporting implausible energy intakes (<800 or >4000 kcal/d in males and <500 or >3500 kcal/d in females) ($n = 660$), or with medical and dietary questionnaires judged as unreliable by the interviewers ($n = 233$ and $n = 939$, respectively). The final sample size available for analysis was of 22,495 participants (Supplemental Figure 1).

Dietary assessment

Food intake during the year before enrolment was assessed by a semiquantitative food frequency questionnaire (FFQ) [24] validated and adapted to the Italian population. The FFQ contains 14 sections [that is, pasta/rice, soup, meat (excluding salami and other cured meats), fish, raw vegetables, cooked vegetables, eggs, sandwiches, salami and other cured meats, cheese, fruit, bread/wine, milk/coffee/cakes, and herbs/spices] with 248 questions concerning 188 different food items. Using a specifically designed software, frequencies and quantities of each food were linked to Italian Food Tables to obtain estimates of daily intake of macro- and micro-nutrients plus energy. The Nova classification [13] was used to categorize each food item into 1 of the following categories according to the extent and purpose of food processing: 1) fresh or minimally processed foods (for example, fruits and vegetables, meat and fish); 2) processed culinary ingredients (for example, honey, butter); 3) processed foods with salt, sugar, or oil (for example, canned or bottled vegetables and legumes, canned fish); 4) UPF (for example, carbonated drinks, processed meat, sweet or savory packaged snacks).

To estimate UPF, we summed up the amount consumed (g/d) of each food group from the fourth category of Nova (a total of 22 foods

and beverages) and calculated the proportion (%) of UPF in the total weight of food and beverages consumed (g/d) by creating a weight ratio. The weight ratio has been suggested to be more appropriate than energy ratio because it better accounts for nonnutritional factors closely coupled to food processing (for example, neo-formed contaminants, additives, and alterations to the food matrix). Participants were then ranked into sex-specific fifths based on the proportion of UPF in the total food eaten. The full list of foods categorized according to the fourth Nova group is available as Supplemental Table 1.

Adherence to the traditional Mediterranean Diet was evaluated by the Mediterranean Diet Score (MDS) developed by Trichopoulou et al. [25]. This score was obtained by assigning 1 point to each consumed food considered healthy (fruits and nuts, vegetables, legumes, fish, cereals; monounsaturated to saturated fats ratio) and whose consumption was above the sex-specific medians of intake in the Moli-sani Study population, free from CVD, cancer, and diabetes; foods presumed to be consumed less frequently in the Mediterranean Diet (that is, meat and dairy products) were scored positively if their consumption was below the median. All other intakes received 0 points. For ethanol, males who consumed 10–50 g/d and females who consumed 5–25 g/d received 1 point; otherwise, the score was 0. The MDS ranged from 0 to 9 (the latter reflecting maximal adherence).

Diet quality was also assessed by the modified Food Standards Agency Nutrient Profiling System (FSAm-NPS) dietary index, which underpins the Nutri-Score front-of-pack labeling system [26]. The FSAm-NPS is a modified version of the nutrient profiling system, which was initially developed in the United Kingdom to regulate television food advertising to children [26]. We calculated the FSAm-NPS score as was previously done in other population cohorts [27], including the Moli-sani Study cohort [15]. For all foods and beverages included in the FFQ, on the basis of composition for each 100 g of content, we allocated 0–40 points for nutrients that should be consumed in limited amounts [A points; that is, total sugars (g), saturated fats (g), sodium (mg), and energy (kJ)] and 0–15 points for nutrients or components that should be promoted [C points; that is, dietary fiber (g), protein (g), and fruit, vegetables, legumes, and nuts (%)]. We calculated the total score of the product by subtracting the sum of C points from the sum of A points. Thus, the final FSAm-NPS score for each food/beverage was based on a scale that could theoretically range from –15 (healthiest food) to 40 (least healthy food).

Ascertainment of covariates

At study entry, information on sociodemographic factors, lifestyles, and clinical variables was obtained by interviewer-administered questionnaires. Personal history of CVD (angina, myocardial infarction, revascularization procedures, peripheral artery diseases, and cerebrovascular events) and cancer was self-reported and confirmed by medical records and therapy. Participants were considered to have hypertension, hyperlipidemia, or diabetes at baseline if they reported having been treated with disease-specific drugs. Leisure-time physical activity was assessed by an interviewer-administered structured questionnaire [28] and expressed as daily energy expenditure in metabolic equivalent task-hours for sport, walking, and gardening. Height and weight were measured, and BMI was calculated as kg/m^2 . Subjects were classified as never-smokers, current smokers, or former smokers (reported not having smoked at all over the previous 12 mo or more). Education was based on the highest qualification attained and was categorized as up to lower secondary (≤ 8 y of study), upper secondary school (8–13 y of study), and postsecondary education (> 13 y of study). Housing tenure was classified as rented, 1 dwelling

ownership, and >1 dwelling ownership. Additional covariates were considered for females only, and included the menopausal status, and use of hormone replacement therapy.

Statistical analysis

Characteristics of the study population overall and across fifths of UPF consumption were presented as absolute count and relevant percentages for categorical variables, or mean values and $\pm$ SD for continuous variables.

A multivariable linear regression analysis (PROC REG in SAS) was used to estimate the relation between UPF consumption (independent variable, modeled as fifths) with Δ age (dependent variable) and results were expressed as regression coefficients (β) and 95% confidence intervals (CIs). Missing values for covariates—that is history of CVD (number of participants with missing data = 362), cancer ($n = 80$), diabetes ($n = 277$), hyperlipidemia ($n = 207$), hypertension ($n = 155$), menopausal status ($n = 12$), education ($n = 16$), housing ($n = 40$), smoking habits ($n = 16$), hormone replacement therapy ($n = 1$), leisure-time physical activity ($n = 192$), and BMI ($n = 13$)—were handled using a multiple imputation technique (SAS PROC MI, followed by PROC MIANALYZE) to maximize data availability for all variables, avoid bias introduced by missing not-at-random data patterns, and achieve robust results over different simulations ($n = 10$ imputed datasets). Covariates were selected based on previous literature and biological plausibility, rather than deferring to statistical criteria. We then fitted a multivariable model controlled for age, sex, energy intake, consumption of alcoholic beverages not included in the UPF Nova Group (that is, wine and beer), education, housing tenure, place of residence, leisure-time physical activity, smoking habit, BMI, CVD, cancer, diabetes, hypertension, hyperlipidemia, aspirin use, menopausal status, and hormone replacement therapy (females only). The association between UPF consumption and biological aging was also investigated using restricted cubic spline functions [29]. Five knots were placed at the 10th, 30th, 50th, 70th, and 90th percentiles to ensure comparability between the analyses using splines and those where estimates were presented across fifths of UPF consumption. Several subgroup analyses were conducted to test the robustness of the findings according to potential effect modification factors: age, sex, BMI groups (normal weight, overweight and obese), smoking habit, and number of comorbidities. Appropriate multiplicative terms for testing interaction were included in the multivariable models to test for a difference of effect of the UPF (modeled as fifths) across subgroups. The multivariable model served as the reference for the mediation analysis used to estimate the contribution of each potential explanatory factor (that is, the MDS or the FSAM-NPS dietary index), which was alternately included into the model. For the mediation analysis, the publicly available % MEDIANTE macro in SAS was used [30] that calculates the point and interval estimates of the proportion explained by 1 or more intermediate variables, with 95% CI and P values. Statistical tests were 2-sided, and P values <0.05 were considered to indicate statistical significance. Data analysis was generated using SAS/STAT software, version 9.4 (SAS Institute Inc.).

Results

The sample under study consisted of 11,726 females (52.0 %) and 10,769 males (48.0%) with a mean CA of 55.6 y (SD $\pm$ 11.7 y; min-max: 34.6–98.7 y), whereas BA was 54.9 y (SD $\pm$ 9.1; min-max:

24.4–107.6 y), and Δ age (=BA – CA) was -0.70 y (SD $\pm$ 7.70; min-max: -40.04 to 54.72). Study participants reported an mean UPF intake of 10.7% (SD $\pm$ 6.7; weight ratio), and the mean energy from UPFs was 18.2% (SD $\pm$ 18.4; energy ratio). Food mostly contributing to the total of UPF consumed in this population were processed meat (17.6%), cakes/pies/pastries (14.2%), and fruit drinks (10.9%) (Supplemental Table 1).

Characteristics of the study population across fifths of UPF consumption are reported in Table 1. Participants consuming more UPF were younger and more educated, more likely to live in urban place, less physically active, and tended to be currently free from chronic diseases, compared with those in the bottom fifth of UPF intake. In multivariable-adjusted regression analyses, highest consumption of UPF (last fifth) was associated with an accelerated biological aging ($\beta = 0.34$; 95% CI: 0.08, 0.61) compared with the lowest category of intake (Table 2).

The spline analysis (Figure 1) showed a P value for the overall association of <0.001 and a P value for nonlinearity = 0.049, indicating a nonlinear relationship between UPF consumption (modeled as a 1% increase in the proportion of UPF in the diet) and biological aging. This relationship becomes more evident at higher levels of UPF intake, particularly in the highest fifth (Figure 1), consistent with the analysis where UPFs were modeled as fifths of consumption (Table 2).

Potential effect modification across predefined subgroups was not statistically supported, although the magnitude of the association between UPF and biological aging (higher compared with lower fifth) tended to be stronger in men than women (Table 2).

The inclusion of the MDS into the multivariable-adjusted regression model only slightly reduced the association of UPF with biological aging by 9.1% (from $\beta = 0.34$ y; 95% CI: 0.08, 0.61 to $\beta = 0.31$ y; 95% CI: 0.04, 0.5559, P value = 0.16), whereas the FSAM-NPS dietary index did not alter this association (Table 3).

Discussion

In this population-based cohort of middle-aged and elderly Italians, a diet rich in UPF (corresponding to >14% of total energy) was associated with accelerated biological aging (Δ age > 0), as measured by a blood-based aging clock, using 36 circulating biomarkers.

Biological aging has been found to be associated with multiple health outcomes in several cohorts [2,7], and could be influenced by lifestyle, including diet [7,31]. In line with other studies [9,10], analyses from the Moli-sani Study cohort have previously reported the beneficial effects of healthy diets, such as the Mediterranean Diet and the DASH diet, on deceleration of biological aging (Δ age < 0) [31]. These favorable effects could be due to the high content in, for example, fruits, vegetables and cereals, which are natural sources of bioactive compounds. Indeed, these may reduce oxidative stress and inflammation, which are relevant to the aging process [32]. Consistently, 2 previous United States observational studies associated higher consumption of UPFs with the lower flavonoid intakes [33], and with lower urinary enterolignan concentrations [34].

However, the issue of food processing is still largely overlooked in the scientific literature, and almost all studies were focused on the poor nutritional content as the privileged mechanism linking highly processed food consumption to health outcomes. Indeed, both in the Seguimiento Universidad de Navarra and the National Health and Nutrition Examination Survey cohorts [16,17], the adverse association of UPF with respective clocks of aging was likely attributed to the high content of unhealthy nutrients characterizing highly processed foods. Similarly, prospective analyses on over 10,000 Brazilian adults

TABLE 1
Main characteristics of the study population by sex-specific fifths of ultra-processed food (UPF) consumption, in the Moli-sani Study cohort (*n* = 22,495).

	All	UPF consumption (weight ratio)				
		Q1	Q2	Q3	Q4	Q5
No. of subjects (%)	22,495	4498 (20.0%)	4499 (20.0%)	4500 (20.0%)	4499 (20.0%)	4499 (20.0%)
UPF (weight ratio; median, min-max)	9.3 (0.0–63.9)	3.9 (0.0–6.0)	6.6 (4.8–8.8)	9.3 (7.1–11.8)	12.8 (9.9–16.0)	19.2 (14.2–63.9)
UPF (energy ratio; median, min-max)	17.3 (0.0–69.9)	8.8 (0.0–26.0)	13.9 (2.6–31.2)	17.7 (4.3–34.8)	21.5 (1.7–45.8)	27.7 (7.7–69.9)
Chronological age; CA (y; means, SD)	55.6 (11.7)	62.0 (10.7)	58.3 (11.3)	55.2 (11.0)	52.5 (11.0)	49.9 (10.8)
Biological age; BA (y; means, SD)	54.9 (9.1)	57.9 (8.6)	55.6 (8.4)	53.9 (8.3)	52.8 (8.2)	51.5 (7.9)
Δage (BA – CA; y; means, SD)	−0.70 (7.7)	−4.1 (8.1)	−2.6 (8.1)	−1.3 (7.9)	0.3 (7.9)	1.6 (7.9)
Males (%)	47.9	47.9	47.9	47.9	47.9	47.9
Education (%)						
Up to lower school	52.4	65.5	57.8	50.9	45.9	41.8
Upper secondary	34.7	25.6	30.3	35.6	39.5	42.7
Postsecondary education	12.8	8.9	11.7	13.4	14.6	15.5
Missing data	0.1	0.02	0.2	0.1	0.0	0.04
Housing tenure (%)						
Rent	8.7	7.1	7.8	8.3	9.3	10.9
One dwelling ownership	82.2	83.6	83.5	82.3	81.5	82.3
>1 dwelling ownership	8.9	9.1	8.5	9.1	9.1	8.7
Missing data	0.2	0.2	0.2	0.3	0.1	0.1
Place of residence (%)						
Urban	67.5	64.7	67.4	66.9	68.5	69.9
Rural	32.5	35.3	32.6	33.1	31.5	30.1
Smoking status (%)						
Nonsmoker	49.6	49.8	49.3	48.6	49.9	50.8
Smokers	22.8	19.8	21.1	22.7	24.4	26.1
Former	27.5	30.3	29.6	28.6	25.9	23.0
Missing data	0.1	0.1	0.0	0.1	0.2	0.1
BMI ¹ (kg/m ² ; means, SD)	28.0 (4.7)	28.7 (4.8)	28.4 (4.7)	28.1 (4.7)	27.6 (4.7)	27.3 (4.8)
Leisure-time PA ² (MET h-d; means, SD)	3.5 (4.0)	3.8 (4.5)	3.6 (4.0)	3.5 (3.8)	3.4 (3.7)	3.3 (3.9)
Menopausal status (%)						
Yes	58.3	79.6	66.8	57.5	48.3	39.0
Missing data	0.1	0.1	0.0	0.1	0.0	0.3
Hormone replacement therapy (%)	5.8	6.7	6.4	6.1	5.5	4.4
Missing data	0.01	0.0	0.0	0.0	0.0	0.0
Cardiovascular disease (%)						
Yes	5.2	4.8	4.3	2.4	2.1	2.6
Missing data	1.6	2.0	1.4	1.7	1.9	1.4
Cancer (%)						
Yes	3.5	5.1	3.9	4.2	3.7	3.1
Missing data	0.4	0.6	0.6	0.3	0.4	0.5
Diabetes (%)						
Yes	4.9	7.5	5.0	2.5	2.3	1.2
Missing data	1.2	1.6	1.4	1.5	1.2	1.5
Hypertension (%)						
Yes	28.6	39.4	33.5	27.9	23.4	17.0
Missing data	0.7	0.9	0.8	0.5	0.5	0.6
Hyperlipidemia (%)						
Yes	7.8	11.8	9.1	6.0	5.0	3.9
Missing data	0.9	1.5	1.4	0.8	0.8	0.6
Aspirin use (%)						
Yes	4.9	5.2	4.3	2.8	2.3	2.2
Missing data	1.8	1.5	2.1	1.1	1.0	1.1

Values are reported as *n* (%) for categorical variables and unadjusted means (SD) for continuous variables. Sex-specific cut-offs for fifths of UPF proportions were 4.8%, 7.1%, 9.9%, and 14.2% in males, and 6.0%, 8.8%, 11.8%, and 16.0% in females.
Abbreviations: MET-h, metabolic equivalent of task hour; PA, physical activity.

¹ Available for 22,482 participants.
² Available for 22,303 participants.

indicated that consumption of UPF >20% of total daily calories was associated with faster rates of both global cognitive and executive function decline, especially among participants with a low healthy diet score [18]. Also, we previously documented a direct association between UPF consumption and subclinical inflammation [35], although the high proinflammatory potential of diets rich in UPF only marginally explained this association; therefore, suggesting that other factors other

than nutrients could link UPFs to inflammation, a condition strictly related to unhealthy aging.
In this study we therefore tested whether the observed association between UPF and accelerated biological aging could be, at least partially, explained by the low nutritional content of these foods, being typically high in sugar, saturated fats, and salt although having low fiber and vitamins content.

TABLE 2
Association¹ of ultra-processed food (UPF) consumption with biological aging (Δ age), in the Moli-sani Study cohort ($n = 22,495$).

	No. of participants	UPF consumption (weight ratio)					<i>P</i> value for interaction ²
		Q1	Q2	Q3	Q4	Q5	
All	22,495	Ref.	−0.15 (−0.39, 0.09)	−0.14 (−0.39, 0.11)	0.10 (−0.16, 0.35)	0.34 (0.08, 0.61)	—
Sex							
Males	10,769	Ref.	−0.02 (−0.38, 0.33)	−0.03 (−0.39, 0.34)	0.18 (−0.20, 0.56)	0.71 (0.31, 1.12)	0.59
Females	11,726	Ref.	−0.25 (−0.58, 0.07)	−0.22 (−0.56, 0.11)	0.08 (−0.26, 0.42)	0.09 (−0.26, 0.44)	
Age groups							
≤ 65 y	17,301	Ref.	−0.22 (−0.50, 0.07)	−0.29 (−0.57, −0.04)	−0.04 (−0.33, 0.25)	0.12 (−0.18, 0.41)	0.95
> 65 y	5194	Ref.	−0.06 (−0.52, 0.39)	0.21 (−0.31, 0.74)	0.04 (−0.54, 0.63)	0.08 (−0.60, 0.76)	
BMI (kg/m ²)							
Normal weight (BMI < 25)	6165	Ref.	−0.14 (−0.65, 0.37)	−0.20 (−0.71, 0.31)	−0.02 (−0.54, 0.49)	0.30 (−0.22, 0.83)	0.41
Overweight (25 ≤ BMI < 30)	9634	Ref.	−0.19 (−0.56, 0.17)	−0.09 (−0.47, 0.28)	0.23 (−0.16, 0.61)	0.22 (−0.19, 0.62)	
Obese (BMI ≥ 30)	6696	Ref.	−0.11 (−0.53, 0.30)	−0.21 (−0.64, 0.22)	0.0004 (−0.46, 0.46)	0.49 (0.0002, 0.97)	
Smoking status							
Nonsmokers	17,351	Ref.	−0.03 (−0.30, 0.24)	−0.17 (−0.45, 0.11)	0.20 (−0.09, 0.49)	0.39 (0.08, 0.69)	0.91
Smokers	5144	Ref.	−0.65 (−1.18, −0.12)	−0.11 (−0.64, 0.42)	−0.29 (−0.83, 0.25)	0.10 (−0.46, 0.65)	
No. of chronic diseases							
None	19,708	Ref.	0.45 (−0.20, 1.10)	0.36 (−0.37, 1.10)	0.27 (−0.52, 1.07)	1.22 (0.34, 2.11)	0.25
1 or more	2787	Ref.	−0.29 (−0.55, −0.03)	−0.26 (−0.53, 0.00002)	0.004 (−0.27, 0.27)	0.18 (−0.10, 0.46)	

¹ Regression coefficients β (95% confidence interval) obtained from a multivariable-adjusted model controlled for age, sex, energy intake, intake of alcoholic beverages not included in the UPF Nova Group (i.e., wine and beer), educational level, housing tenure, place of residence, leisure-time physical activity, smoking habit, BMI, history of cardiovascular disease, cancer, diabetes, hypertension, hyperlipidemia, aspirin use, menopausal status, and hormone replacement therapy (females only).

² Terms of interaction obtained by modeling UPF consumption as fifths.

Our results showed that the UPFs-biological aging association was weakly explained by the poor nutritional composition of these highly processed foods, suggesting that biological aging could be mainly influenced by nonnutrient food characteristics, which include altered food matrix, contact materials and neo-formed compounds.

UPFs are a source of acrylamide, a chemical that naturally forms in starchy food products during high-temperature cooking, including

industrial processing [36]. In the United States population, greater UPF consumption has been associated with increased acrylamide exposure [37]. Acrylamide is classified as neurotoxic with potential negative effects on the cognitive function [36]. Indeed, high levels of acrylamide have been associated with increased oxidative stress and inflammation [38], whereas chronic exposure in vitro accelerates senescence, which contributes to endothelial aging [39].

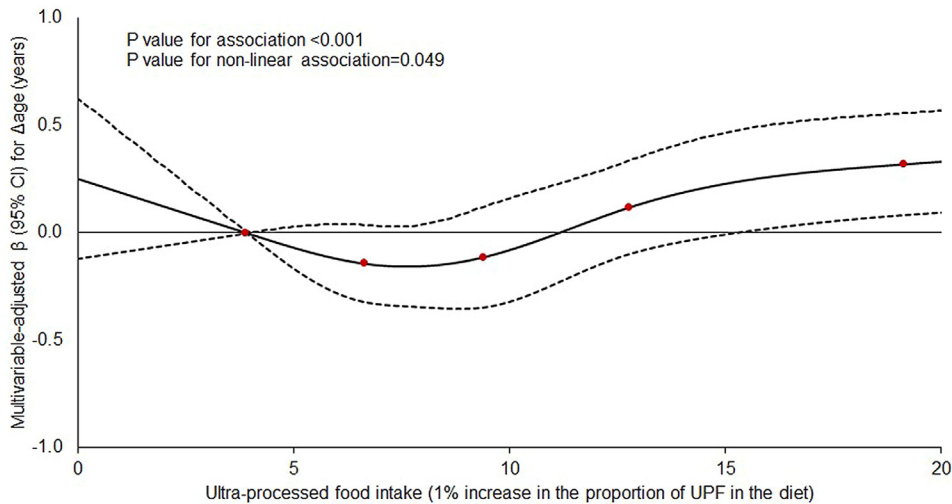


FIGURE 1. Multivariable dose–response association of biological aging with ultra-processed food consumption (1% increase in the proportion of ultra-processed food in the diet) in the Moli-sani Study cohort ($n = 22,495$). Regression coefficients β with 95% confidence intervals were obtained from the multivariable model adjusted for age, sex, energy intake, intake of alcoholic beverages not included in the Nova Group 4 (i.e., wine and beer), educational level, housing tenure, place of residence, leisure-time physical activity, smoking habit, BMI, history of cardiovascular disease, cancer, diabetes, hypertension, hyperlipidemia, aspirin use, menopausal status, and hormone replacement therapy (women only). Ultra-processed food consumption was modeled as a continuous exposure, and the 10th percentile served as the reference value (3.9% of ultra-processed food consumption; weight ratio). The dashed lines indicate 95% confidence bands. Five knots were placed at the 10th, 30th, 50th, 70th, and 90th percentiles.

TABLE 3
Mediation analysis testing overall diet quality as an explanatory factor of the association between ultra-processed food (UPF; weight ratio) consumption and biological aging (Δ age) in the Moli-sani Study cohort ($n = 22,495$).

	UPF consumption (fifths of) ¹				
	Q1	Q2	Q3	Q4	Q5
UPF + Mediterranean diet score	Reference	−0.15 (−0.39, 0.09)	−0.14 (−0.39, 0.11)	0.10 (−0.16, 0.35)	0.34 (0.08, 0.61)
Proportion explained ² (95% CI); <i>P</i> value	—	NM	NM	20.4 (0.4, 94.2); 0.16	9.1 (1.0, 50.9); 0.16
UPF + FSAm-NPS dietary index	Reference	−0.14 (−0.38, 0.11)	−0.12 (−0.37, 0.13)	0.13 (−0.13, 0.38)	0.38 (0.10, 0.65)
Proportion explained ² (95% CI); <i>P</i> value	—	10.7 (0.8, 64.1); 0.14	16.1 (1.0, 79.1); 0.14	NM	NM

Abbreviations: FSAm-NPS, modified Food Standards Agency Nutrient Profiling System; NM, not mediating.

¹ Regression coefficients β [95% confidence interval (CI)] obtained from a multivariable model controlled for age, sex, energy intake, intake of alcoholic beverages not included in the Nova Group 4 (i.e., wine and beer), educational level, housing tenure, place of residence, leisure-time physical activity, smoking habit, BMI, history of cardiovascular disease, cancer, diabetes, hypertension, hyperlipidemia, aspirin use, menopausal status, and hormone replacement therapy (females only).

² Proportion of effect explained by intermediate variables (i.e., diet quality scores) with 95% CI and relevant *P* values as produced by the %MEDIATE macro in SAS.

Food processing can produce other potentially toxic compounds such as furans, heterocyclic amines, polycyclic aromatic hydrocarbons, acrolein, industrial trans-fatty acids, and advanced glycation end products [40]; accumulation of advanced glycation end products may alter the structure and function of proteins, consequently affecting several hallmarks of aging [41].

Also, UPFs are usually wrapped in plastic that is a source of bisphenols and phthalates, which are multifunctional synthetic chemicals used to make plastics flexible and durable, and exert endocrine disruptor actions [40]. United States observational studies have associated greater consumption of UPFs with higher urinary concentrations of several phthalates and bisphenols [42,43]; many non-UPF products nowadays are also wrapped in plastic; migration of these chemicals into food could be more accentuated in UPF due to their extended shelf life.

Epidemiological studies reported that exposure to bisphenols and phthalates is associated with oxidative stress and insulin resistance in the elderly [44], and may lead to dysregulated inflammatory responses [45]. High concentrations of phthalates have been also linked to the decline of lung function that is found to be partially related to shorter telomere length, a mechanism that contributes to cellular aging and age-related disease [46].

These observations are well substantiated by several in vitro and animal studies demonstrating that oxidative stress and inflammation could be important pathogenic mechanisms triggered by exposure to plasticizers [47].

Additionally, UPFs are characterized by the deconstruction of the food matrix, which is the “architecture” of the food that stems from nutrient interactions and determines very important nutritional properties, including glycemic responses, and nutrient bioavailability [12]. Modifications to the physical structure of the food matrix during processing, as well as usage of food additives, including emulsifiers and nonnutritive sweeteners, may alter gut microbiota composition and function [40], and impair glucose tolerance [48], and this may trigger inflammatory processes, and increase susceptibility to age-related chronic diseases [49].

Strengths and limitations

To the best of our knowledge, this is possibly the first study that examined the association between UPF consumption (according to the Nova classification) and biological aging using circulating blood biomarkers. Major strengths of this study include the use of a well-established population-based cohort, and a large number of

circulating biomarkers used to compute BA. Also, we relied on a large set of covariates to minimize confounding.

However, our study has several limitations too. First, the cross-sectional design of the study makes it difficult to define the temporal directionality of the association; also, observational studies do not allow for causal inference. Second, reverse causation (which is uncontrolled for in cross-sectional designs) might dilute the studied association toward the null; moreover, residual confounding might explain part of the observed associations. Third, the dietary data were collected using an FFQ based on self-reported information that may lead to recall bias and difficulties in assessing portion size. However, these problems may be partially mitigated by exclusion of participants with implausible energy intakes and energy adjustment. Fourth, the FFQ used in this study was not originally developed to assess the degree of food processing or the consumption of UPFs according to the Nova classification and therefore did not include many UPF food items (for example, pre-prepared dishes, energy bars, slimming products); this could have led to a nondifferential underestimation of UPF intake biasing the studied association with biological aging toward the null. Yet, most existing large cohort studies examining the association between UPF consumption and human health relied on the use of FFQs [50]. Fifth, other clocks of aging could be more accurate and robust than ours, but this blood-based measure of biological aging could be used to screen health status and mortality risk, through the use of cheap and easy to obtain medical health records [51]. Finally, caution is needed in generalizing these findings to other populations.

In conclusion cross-sectional analyses from the Moli-sani Study cohort in Italy revealed an association between a high consumption of UPFs and blood-based biological aging, as already observed in few other cohort studies using different aging clocks [16–19]. Our data also shed light on the potential mechanisms involved in such relationship, showing that the poor nutritional content could be only a minor pathway through which UPFs may be adversely associated with the aging process. Further investigations on the nonnutrient components of highly processed foods are therefore warranted.

Acknowledgments

We are grateful to Moli-sani Study participants who enthusiastically joined the study, and to the Associazione Cuore Sano ETS (Campobasso, Italy) for its support to the research communication activities.

ER was supported by the Fondazione Umberto Veronesi, which is gratefully acknowledged. The enrolment phase of the Moli-sani Study was supported by research grants from Pfizer Foundation (Rome, Italy), the Italian Ministry of University and Research (MIUR, Rome, Italy)–Programma Triennale di Ricerca, decreto No1588, and Instrumentation Laboratory, Milan, Italy. The authors would like to thank the colleagues of the Neuromed Biobanking Center for management of the samples and valuable support and cooperation.

Author contributions

The authors' responsibilities were as follows – LI, MB, SE, AG: conceived the study, contributed to its design, and to interpretation of data; SC, AG, ER, MP: managed data collection; ADeC: performed laboratory tests; ADiC, SE AG, AP: analyzed the data; SE, MB: wrote the manuscript; MBD, CC, GdG, LI: originally inspired the Moli-sani Study and critically reviewed the manuscript; and all authors: gave final approval and agree to be accountable for all aspects of the work ensuring its integrity and accuracy.

Conflict of interest

The authors declare no conflict of interest.

Funding

This article was developed within the project funded by Next Generation EU – “Age-It – Ageing well in an ageing society” project (PE0000015), National Recovery and Resilience Plan (NRRP) – PE8 – Mission 4, C2, Intervention 1.3.” The views and opinions expressed are only those of the authors and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the European Commission can be held responsible for them. These analyses were partially supported by the Italian Ministry of Health (Ricerca Corrente 2022–2024). AG was funded by the Italian Ministry of Health (Ricerca Finalizzata U40, project GR-2021-12375341). The funders had no role in planning the study design or in the collection, analysis, or interpretation of data, the writing of the report, or the decision to submit the article for publication. All authors were and are independent from the funders.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author. The data are stored in an institutional repository (<https://repository.neuromed.it>), and access is restricted by the ethical approvals and the legislation of the European Union.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ajcnut.2024.10.006>.

References

- [1] D. Marsman, D.W. Belsky, D. Gregori, M.A. Johnson, T. Low Dog, S. Meydani, et al., Healthy ageing: the natural consequences of good nutrition-a conference report, *Eur. J. Nutr.* 57 (Suppl 2) (2018) 15–34, <https://doi.org/10.1007/s00394-018-1723-0>.
- [2] M.R. Hameczyk, R.M. Nevado, A. Baretino, V. Fuster, V. Andrés, Biological versus chronological aging: JACC focus seminar, *J. Am. Coll. Cardiol.* 75 (8) (2020) 919–930, <https://doi.org/10.1016/j.jacc.2019.11.062>.
- [3] P. Mamoshina, K. Kochetov, E. Putin, F. Cortese, A. Aliper, W.S. Lee, et al., Population specific biomarkers of human aging: a big data study using South Korean, Canadian, and Eastern European patient populations, *J. Gerontol. A Biol. Sci. Med. Sci.* 73 (11) (2018) 1482–1490, <https://doi.org/10.1093/gerona/gly005>.
- [4] A. Gialluisi, A. Santoro, A. Tirozzi, C. Cerletti, M.B. Donati, G. de Gaetano, et al., Epidemiological and genetic overlap among biological aging clocks: new challenges in biogerontology, *Ageing Res. Rev.* 72 (2021) 101502, <https://doi.org/10.1016/j.arr.2021.101502>.
- [5] T.V. Pyrkov, P.O. Fedichev, Biological age is a universal marker of aging, stress, and frailty, *BioRxiv* 10 (2019) 578245.
- [6] G. Passarino, F. De Rango, A. Montesanto, Human longevity: genetics or lifestyle? It takes two to tango, *Immun. Ageing* 13 (2016) 12, <https://doi.org/10.1186/s12979-016-0066-z>.
- [7] A. Gialluisi, A. Di Castelnuovo, S. Costanzo, M. Bonaccio, M. Persichillo, S. Magnacca, et al., Exploring domains, clinical implications and environmental associations of a deep learning marker of biological ageing, *Eur. J. Epidemiol.* 37 (1) (2022) 35–48, <https://doi.org/10.1007/s10654-021-00797-7>.
- [8] World Health Organization, Ageing and health .2022 [cited July 2024]. Available from: <https://www.who.int/news-room/fact-sheets/detail/ageing-and-health>.
- [9] O.M. Shannon, A.W. Ashor, F. Scialo, G. Saretzki, C. Martin-Ruiz, J. Lara, et al., Mediterranean diet and the hallmarks of ageing, *Eur. J. Clin. Nutr.* 75 (8) (2021) 1176–1192, <https://doi.org/10.1038/s41430-020-00841-x>.
- [10] L. Huang, Y. Tao, H. Chen, X. Chen, J. Shen, C. Zhao, et al., Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay (MIND) diet and cognitive function and its decline: a prospective study and meta-analysis of cohort studies, *Am. J. Clin. Nutr.* 118 (1) (2023) 174–182, <https://doi.org/10.1016/j.ajcnut.2023.04.025>.
- [11] C.F. Martínez, S. Esposito, A. Di Castelnuovo, S. Costanzo, E. Ruggiero, A. De Curtis, et al., Association between the inflammatory potential of the diet and biological aging: a cross-sectional analysis of 4510 adults from the Moli-sani study cohort, *Nutrients* 15 (6) (2023) 1503, <https://doi.org/10.3390/nu15061503>.
- [12] A. Fardet, E. Rock, Chronic diseases are first associated with the degradation and artificialization of food matrices rather than with food composition: calorie quality matters more than calorie quantity, *Eur. J. Nutr.* 61 (5) (2022) 2239–2253, <https://doi.org/10.1007/s00394-021-02786-8>.
- [13] C.A. Monteiro, Nutrition and health: the issue is not food, nor nutrients, so much as processing, *Public Health Nutr* 12 (5) (2009) 729–731, <https://doi.org/10.1017/S1368980009005291>.
- [14] C.A. Monteiro, G. Cannon, R. Levy, J.C. Moubarac, P. Jaime, A.P. Martins, et al., NOVA. The star shines bright. Food classification, *Public Health World Nutr* 7 (2016) 28–38.
- [15] M. Bonaccio, A. Di Castelnuovo, E. Ruggiero, S. Costanzo, G. Grosso, A. De Curtis, et al., Joint association of food nutritional profile by Nutri-Score front-of-pack label and ultra-processed food intake with mortality: Moli-sani prospective cohort study, *BMJ* 378 (2022) e070688, <https://doi.org/10.1136/bmj-2022-070688>.
- [16] L. Alonso-Pedrero, A. Ojeda-Rodríguez, M.A. Martínez-González, G. Zalba, M. Bes-Rastrollo, A. Marti, Ultra-processed food consumption and the risk of short telomeres in an elderly population of the Seguimiento Universidad de Navarra (SUN) Project, *Am. J. Clin. Nutr.* 111 (6) (2020) 1259–1266, <https://doi.org/10.1093/ajcn/nqaa075>.
- [17] B.R. Cardoso, P. Machado, E.M. Steele, Association between ultra-processed food consumption and cognitive performance in US older adults: a cross-sectional analysis of the NHANES 2011–2014, *Eur. J. Nutr.* 61 (8) (2022) 3975–3985, <https://doi.org/10.1007/s00394-022-02911-1>.
- [18] N. Gomes Gonçalves, N. Vidal Ferreira, N. Khandpur, E. Martinez Steele, R. Bertazzi Levy, P. Andrade Lotufo, et al., Association between consumption of ultraprocessed foods and cognitive decline, *JAMA Neurol* 80 (2) (2023) 142–150, <https://doi.org/10.1001/jamaneurol.2022.4397>.
- [19] H. Sandoval-Insausti, R. Blanco-Rojo, A. Graciani, E. López-García, B. Moreno-Franco, M. Laclaustra, et al., Ultra-processed food consumption and incident frailty: a prospective cohort study of older adults, *J. Gerontol. A Biol. Sci. Med. Sci.* 75 (6) (2020) 1126–1133, <https://doi.org/10.1093/gerona/glz140>.
- [20] S. Carlos, C. De La Fuente-Arrillaga, M. Bes-Rastrollo, C. Razquin, A. Rico-Campà, M.A. Martínez-González, et al., Mediterranean diet and health outcomes in the SUN cohort, *Nutrients* 10 (4) (2018) 439, <https://doi.org/10.3390/nu10040439>.
- [21] L. Allen, C. Sudlow, P. Downey, T. Peakman, J. Danesh, P. Elliott, et al., UK Biobank: current status and what it means for epidemiology, *Health Policy Technol* 1 (3) (2012) 123–126, <https://doi.org/10.1016/j.hlpt.2012.07.003>.
- [22] L. Iacoviello, A. Bonanni, S. Costanzo, A. De Curtis, A. Di Castelnuovo, M. Olivieri, et al., The Moli-Sani project, a randomized, prospective cohort study in the Molise region in Italy; design, rationale and objectives, *Italian J. Public Health.* 4 (2) (2007) 110–118.

- [23] T. Zeller, M. Hughes, T. Tuovinen, et al., BiomarCaRE: rationale and design of the European BiomarCaRE project including 300,000 participants from 13 European countries, *Eur. J. Epidemiol.* 29 (10) (2014) 777–790, <https://doi.org/10.1007/s10654-014-9952-x>.
- [24] P. Pisani, F. Faggiano, V. Krogh, D. Palli, P. Vineis, F. Berrino, Relative validity and reproducibility of a food frequency dietary questionnaire for use in the Italian EPIC centres, *Int. J. Epidemiol.* 26 (Suppl 1) (1997) S152–S160, https://doi.org/10.1093/ije/26.suppl_1.s152.
- [25] A. Trichopoulou, T. Costacou, C. Bamia, D. Trichopoulos, Adherence to a Mediterranean diet and survival in a Greek population, *N. Engl. J. Med.* 348 (26) (2003) 2599–2608, <https://doi.org/10.1056/NEJMoa025039>.
- [26] C. Julia, S. Hercberg, Development of a new front-of-pack nutrition label in France: the five-Colour Nutri-Score, *Public Health Panorama* 3 (2017) 712–725.
- [27] M. Deschasaux, I. Huybrechts, C. Julia, S. Hercberg, M. Egnell, B. Srour, et al., Association between nutritional profiles of foods underlying Nutri-Score front-of-pack labels and mortality: EPIC cohort study in 10 European countries, *BMJ* 370 (2020) m3173, <https://doi.org/10.1136/bmj.m3173>.
- [28] A. Mannocci, D. Di Thiene, A. Del Cimmuto, D. Masala, A. Boccia, E. De Vito, et al., International physical activity questionnaire: validation and assessment in an Italian sample, *Ital. J. Public Health.* 7 (4) (2010) 369–376.
- [29] L. Desquilbet, F. Mariotti, Dose-response analyses using restricted cubic spline functions in public health research, *Stat. Med.* 29 (9) (2010) 1037–1057.
- [30] E. Hertzmark, M. Pazaris, D. Spiegelman, The SAS MEDIAN macro, Harvard T.H. Chan School of Public Health, Boston, 2012.
- [31] S. Esposito, A. Gialluisi, S. Costanzo, A. Di Castelnuovo, E. Ruggiero, A. De Curtis, et al., Mediterranean diet and other dietary patterns in association with biological aging in the Moli-sani Study cohort, *Clin. Nutr.* 41 (5) (2022) 1025–1033, <https://doi.org/10.1016/j.clnu.2022.02.023>.
- [32] A. Serino, G. Salazar, Protective role of polyphenols against vascular inflammation, aging and cardiovascular disease, *Nutrients* 11 (1) (2018) 53, <https://doi.org/10.3390/nu11010053>.
- [33] A.E. Leitão, H. Roschel, G. Oliveira-Júnior, R. Genario, T. Franco, C.A. Monteiro, et al., Association between ultra-processed food and flavonoid intakes in a nationally representative sample of the US population, *Br. J. Nutr.* 131 (6) (2024) 1074–1083, <https://doi.org/10.1017/S00071145230002568>.
- [34] E. Martínez Steele, C.A. Monteiro, Association between dietary share of ultra-processed foods and urinary concentrations of phytoestrogens in the US, *Nutrients* 9 (3) (2017) 209, <https://doi.org/10.3390/nu9030209>.
- [35] C. Mignogna, S. Costanzo, A. Di Castelnuovo, E. Ruggiero, N. Shivappa, J.R. Hebert, et al., The inflammatory potential of the diet as a link between food processing and low-grade inflammation: an analysis on 21,315 participants to the Moli-sani study, *Clin. Nutr.* 41 (10) (2022) 2226–2234, <https://doi.org/10.1016/j.clnu.2022.08.020>.
- [36] Z.M. Liu, L.A. Tse, B. Chen, S. Wu, D. Chan, T. Kowk, et al., Dietary acrylamide exposure was associated with mild cognition decline among non-smoking Chinese elderly men, *Sci. Rep.* 7 (1) (2017) 6395, <https://doi.org/10.1038/s41598-017-06813-9>.
- [37] E. Martínez Steele, J.P. Buckley, C.A. Monteiro, Ultra-processed food consumption and exposure to acrylamide in a nationally representative sample of the US population aged 6 years and older, *Prev. Med.* 174 (2023) 107598, <https://doi.org/10.1016/j.ypmed.2023.107598>.
- [38] B.P. Riboldi, Á.M. Vinhas, J.D. Moreira, Risks of dietary acrylamide exposure: a systematic review, *Food Chem* 157 (2014) 310–322, <https://doi.org/10.1016/j.foodchem.2014.02.046>.
- [39] C. Sellier, E. Boulanger, F. Maladry, F.J. Tessier, R. Lorenzi, R. Nevière, et al., Acrylamide induces accelerated endothelial aging in a human cell model, *Food Chem Toxicol* 83 (2015) 140–145, <https://doi.org/10.1016/j.fct.2015.05.021>.
- [40] B. Srour, M.C. Kordahi, E. Bonazzi, M. Deschasaux-Tanguy, M. Touvier, B. Chassaing, Ultra-processed foods and human health: from epidemiological evidence to mechanistic insights, *Lancet Gastroenterol Hepatol* 7 (12) (2022) 1128–1140, [https://doi.org/10.1016/S2468-1253\(22\)00169-8](https://doi.org/10.1016/S2468-1253(22)00169-8).
- [41] J. Chaudhuri, Y. Bains, S. Guha, A. Kahn, D. Hall, N. Bose, et al., The role of advanced glycation end products in aging and metabolic diseases: bridging association and causality, *Cell Metab.* 28 (3) (2018) 337–352, <https://doi.org/10.1016/j.cmet.2018.08.014>.
- [42] J.P. Buckley, H. Kim, E. Wong, C.M. Rebholz, Ultra-processed food consumption and exposure to phthalates and bisphenols in the US National Health and Nutrition Examination Survey, 2013–2014, *Environ. Int.* 131 (2019) 105057.
- [43] E. Martínez Steele, N. Khandpur, M.L. da Costa Louzada, C.A. Monteiro, Association between dietary contribution of ultra-processed foods and urinary concentrations of phthalates and bisphenol in a nationally representative sample of the US population aged 6 years and older, *PLOS ONE* 15 (7) (2020) e0236738, <https://doi.org/10.1371/journal.pone.0236738>.
- [44] J.H. Kim, H.Y. Park, S. Bae, Y.H. Lim, Y.C. Hong, Diethylhexyl phthalates is associated with insulin resistance via oxidative stress in the elderly: a panel study, *PLOS ONE* 8 (8) (2013) e71392, <https://doi.org/10.1371/journal.pone.0071392>.
- [45] Z. Liu, Y. Lu, K. Zhong, C. Wang, X. Xu, The associations between endocrine disrupting chemicals and markers of inflammation and immune responses: a systematic review and meta-analysis, *Ecotoxicol. Environ. Saf.* 234 (2022) 113382, <https://doi.org/10.1016/j.ecoenv.2022.113382>.
- [46] J. Hou, W. Yin, P. Li, C. Hu, T. Xu, J. Cheng, et al., Joint effect of polycyclic aromatic hydrocarbons and phthalates exposure on telomere length and lung function, *J. Hazard Mater.* 386 (2020) 121663, <https://doi.org/10.1016/j.jhazmat.2019.121663>.
- [47] A. Brulport, C. Lencina, M.C. Chagnon, L. Le Corre, L. Guzylack-Piriou, Transgenerational effects on intestinal inflammation status in mice perinatally exposed to bisphenol S, *Chemosphere* 262 (2021) 128009, <https://doi.org/10.1016/j.chemosphere.2020.128009>.
- [48] J. Suez, Y. Cohen, R. Valdés-Mas, U. Mor, M. Dori-Bachash, S. Federici, et al., Personalized microbiome-driven effects of non-nutritive sweeteners on human glucose tolerance, *Cell* 185 (18) (2022) 3307–3328.e19, <https://doi.org/10.1016/j.cell.2022.07.016>.
- [49] S. Donati Zeppa, D. Agostini, F. Ferrini, M. Gervasi, E. Barbieri, A. Bartolacci, et al., Interventions on gut microbiota for healthy aging, *Cells* 12 (1) (2022) 34, <https://doi.org/10.3390/cells12010034>.
- [50] M.M. Lane, E. Gamage, S. Du, D.N. Ashtree, A.J. McGuinness, S. Gauci, et al., Ultra-processed food exposure and adverse health outcomes: umbrella review of epidemiological meta-analyses, *BMJ* 384 (2024) e077310, <https://doi.org/10.1136/bmj-2023-077310>.
- [51] A. Gialluisi, A. Di Castelnuovo, M.B. Donati, G. de Gaetano, L. Iacoviello, Moli-sani Study Investigators, Machine learning approaches for the estimation of biological aging: the road ahead for population studies, *Front Med. (Lausanne)* 6 (2019) 146, <https://doi.org/10.3389/fmed.2019.00146>.